XRPL Validator — Daily Engineering Report · 2026-11-10

Period: 2026-11-10 00:00 to 24:00 US Eastern (2026-11-10 05:00 to 2026-11-11 05:00 UTC); ledgers #108,232,393 to #108,253,669. Every count from the ledger and the validator's logs stops at #108,253,669, closed 04:59:51 UTC. Generated 2026-11-11 00:20 EST by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Degraded — 1 engine receipt open.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,275 of the 21,277 ledgers the network closed (99.991%); every validation matched the network's validated ledger. Unsigned: 0 during planned restarts, 2 outside maintenance (0 incidents, 1 isolated miss).
- **Native engine:** 1 engine receipt: #108,241,775 — triaged: an isolated replay reproduced the difference.
- **Changes:** 0 deployments, 0 commits merged into main.
- **Network:** 17 slow-close periods; the longest, 15:26 UTC–15:28 UTC, closed ledgers up to 20 s apart.
- **Independence program:** 13 of 30 qualifying days (snapshot at 05:20:35 UTC).

Service levels

Measure	Value
Ledgers closed by the network	21,277
Validations issued	21,275
Coverage (validations / ledgers closed)	99.991%
Validations matching the network's validated ledger hash	21,275 of 21,275
Unsigned during planned restarts	0
Unsigned outside maintenance	2
Coverage outside planned maintenance (objective 99.9%)	99.991%
Restarts	0
Basis: our own ledger hash	21,270
Basis: our own hash, witnessed by the reference node (quorum short)	5
External agreement, data.xrpl.org (UTC day 2026-11-10)	0.99985 (3 missed of 20,408)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#108,232,393–#108,253,669), by step:

Step	Ledgers with a receipt
Step 1 · native engine	1
Step 2 · records	1
Step 3 · order	1
Step 4 · peer data	0
Step 4 · validations	7
Step 5 · own hash	1
Writer	1

Live counters, snapshot at 05:20:35 UTC (since the last restart, 2026-11-05 00:49 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	130,932 ledgers, 27,418,796 transactions	130,930 fully matched; results 2, bytes 64, keys 1/12
2 · transaction records	132,972 ledgers, 27,951,149 records	27,950,889 exact; differ 256, tree 3
3 · transaction order	133,015 ledgers (mode: on)	133,010 exact; order 5, results 1, bytes 0, refused 5
4 · ledger data from peers	133,021 ledgers fetched	0 matched; differ 0, failed 0
4 · trusted validations	133,000 ledgers judged	132,967 agreed; differ 0, short of quorum 33
5 · our own ledger hash	133,019 ledgers	132,963 own; differ 3, partial 53
Writer	every ledger written	mismatch 0, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	15:26 UTC	#108,241,775	Step 1 · native engine, Step 2 · records, Step 3 · order, Step 5 · own hash, Writer	Payment tx 9F6C18A1A388: RippleState 12CB8DFB (modified on the network): FinalFields.Balance ours= {"currency":"524C55534400000000000…	triaged: an isolated replay reproduced the difference; no fix recorded for this ledger
R-02	15:26 UTC	#108,241,775–#108,241,778 (4)	Step 4 · validations	16–18 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (2 of 4 signed)
R-03	16:27 UTC	#108,242,551–#108,242,553 (3)	Step 4 · validations	18–24 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network

ID	Time	Ledger(s)	Step	Description	Disposition
					validated these ledgers normally (all signed)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-11-10-01, INC-01 is INC-2026-11-10-01.

Incidents

No incidents.

Isolated unsigned ledgers (shorter than 3 ledgers, cause not recorded): #108,241,775–#108,241,776 (2) at 15:26:21 UTC.

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

Qualifying days: **13 of 30** (snapshot at 05:20:35 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-11-10 — tracked #108,227,899–#108,249,199	Value
Validations	21,299 (21,299 own, 0 borrowed)
Coverage of the tracked ledgers	99.99%
Our own transaction order	on
Ledger data from peers	yes

Meets every criterion.

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,277 (#108,232,393–#108,253,669)
Mean close interval	4.06 s
Longest close interval	39 s (before #108,242,552)
Closes at least 15 s apart	27
Empty ledgers	0

- Slow closes 06:01:51 UTC–06:02:10 UTC: ledgers #108,233,320, 1 close at least 15 s apart, up to 19 s.
- Slow closes 13:31:32 UTC–13:31:50 UTC: ledgers #108,240,133, 1 close at least 15 s apart, up to 18 s.
- Slow closes 13:53:41 UTC–13:55:00 UTC: ledgers #108,240,456–#108,240,461 (6), 3 closes at least 15 s apart, up to 39 s.
- Slow closes 14:34:32 UTC–14:35:00 UTC: ledgers #108,241,047, 1 close at least 15 s apart, up to 28 s.
- Slow closes 15:21:41 UTC–15:22:00 UTC: ledgers #108,241,728, 1 close at least 15 s apart, up to 19 s.
- Slow closes 15:24:01 UTC–15:24:20 UTC: ledgers #108,241,750, 1 close at least 15 s apart, up to 19 s.
- Slow closes 15:26:21 UTC–15:28:00 UTC: ledgers #108,241,776–#108,241,782 (7), 5 closes at least 15 s apart, up to 20 s.
- Slow closes 15:45:31 UTC–15:45:50 UTC: ledgers #108,241,970, 1 close at least 15 s apart, up to 19 s.
- Slow closes 16:27:41 UTC–16:29:00 UTC: ledgers #108,242,552–#108,242,554 (3), 3 closes at least 15 s apart, up to 39 s.
- Slow closes 16:29:03 UTC–16:29:20 UTC: ledgers #108,242,558, 1 close at least 15 s apart, up to 17 s.
- Slow closes 16:37:51 UTC–16:38:10 UTC: ledgers #108,242,673, 1 close at least 15 s apart, up to 19 s.
- Slow closes 19:08:22 UTC–19:08:40 UTC: ledgers #108,244,839, 1 close at least 15 s apart, up to 18 s.
- Slow closes 21:00:50 UTC–21:01:10 UTC: ledgers #108,246,514, 1 close at least 15 s apart, up to 20 s.
- Slow closes 21:27:52 UTC–21:28:30 UTC: ledgers #108,246,910–#108,246,911 (2), 2 closes at least 15 s apart, up to 20 s.
- Slow closes 00:29:42 UTC–00:30:00 UTC: ledgers #108,249,655, 1 close at least 15 s apart, up to 18 s.
- Slow closes 01:55:00 UTC–01:55:20 UTC: ledgers #108,250,918, 1 close at least 15 s apart, up to 20 s.
- Slow closes 01:55:31 UTC–01:56:20 UTC: ledgers #108,250,922–#108,250,923 (2), 2 closes at least 15 s apart, up to 29 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

No amendment held a majority.

Amendments enabled on mainnet: 98.

Open items

- Receipt R-01 (#108,241,775): open, under investigation.
- Unsigned #108,241,775–#108,241,776 (2) (15:26:21 UTC): cause not recorded.

Corrections to earlier reports

None.